Electricity Consumption Analysis and Prediction in India (2019–2024): A Comprehensive Data Science Study Using Statistical Analysis and Machine Learning

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Abstract—India's rapidly expanding economy and population have generated unprecedented pressure on its national electricity infrastructure, making accurate consumption analysis and generation forecasting of critical strategic importance. This paper presents a rigorous, end-to-end data science study of India's monthly electricity generation spanning January 2019 to November 2025, utilizing the Ember Climate long-format dataset comprising 238,200 records across 13 variables. The analytical pipeline encompasses data ingestion from CSV and SQLite databases, preprocessing with Pandas and NumPy, and multi-modal visualization using Matplotlib and Seaborn. The dataset, filtered for national-level total generation in gigawatt-hours (GWh), yields 83 monthly observations with a mean of 132,204.76 GWh and a maximum of 167,414.52 GWh recorded in June 2024. Exploratory data analysis reveals a strong positive correlation (r = 0.84) between Year and Value, confirming a consistent upward generation trend. A year-wise boxplot analysis demonstrates annual median growth from 115,949 GWh in 2019 to 150,523 GWh in 2024, interrupted only by a measurable contraction in 2020 attributable to the COVID-19 national lockdown. Statistical hypothesis testing using the Shapiro-Wilk, independent t-test, and chi-square procedures validates the observed seasonal and temporal patterns. Coal remains the dominant generation source while Solar exhibits the steepest growth trajectory. No statistically significant outliers were detected using the IQR method. The findings provide actionable insights for grid planners, renewable energy investors, and policymakers seeking data-driven evidence for India's energy transition.

Keywords—*Electricity Consumption; Machine Learning; Exploratory Data Analysis; Energy Forecasting; Linear Regression; Statistical Analysis; India; Ember Climate Dataset*

I. INTRODUCTION

India stands as the world's third-largest electricity consumer, with demand projected to more than double by 2040 relative to 2019 levels according to the International Energy Agency [1]. The nation's dual imperative—meeting rapidly growing energy demand while simultaneously transitioning to cleaner fuel sources—requires evidence-based decision-making supported by rigorous analytical methods. The proliferation of high-resolution monthly electricity datasets, made publicly available by organizations such as Ember Climate, has created an unprecedented opportunity to apply data science techniques to real-world energy challenges.

Traditional approaches to energy planning relied heavily on deterministic engineering models and expert extrapolation. The past decade has witnessed a fundamental shift toward data-driven methodologies, with machine learning and statistical analysis now embedded in grid planning, demand forecasting, and policy evaluation processes worldwide [2]. Despite this growing literature, comprehensive studies applying the full data science lifecycle to India's national monthly electricity data remain scarce.

This paper addresses that gap by systematically applying six analytical stages—data collection, preprocessing, visualization, exploratory analysis, statistical testing, and machine learning—to the Ember Climate India electricity dataset. The dataset spans January 2019 to November 2025, capturing six complete years of monthly generation across all major fuel types including coal, gas, hydro, nuclear, solar, and wind, organized by state and national aggregate levels.

The primary objectives of this study are: (i) to characterize temporal and seasonal trends in India's electricity generation from 2019 to 2024; (ii) to perform formal statistical hypothesis tests validating observed patterns; (iii) to develop and evaluate machine learning models for generation forecasting; and (iv) to derive policy-relevant insights from the analytical findings.

II. LITERATURE REVIEW

Suganthi and Samuel conducted a seminal review of energy demand forecasting models, identifying regression analysis, time series methods, and neural networks as the three dominant approaches [3]. Their analysis found that hybrid models combining statistical and computational intelligence components consistently outperformed single-method approaches in medium-term forecasting horizons, providing the foundational motivation for the multi-technique framework employed in the present study.

Rallapalli and Ghosh applied multiple linear regression and autoregressive integrated moving average (ARIMA) models to forecast India's electricity demand through 2020, demonstrating that year, installed capacity, and GDP are the strongest linear predictors of aggregate demand [4]. Their findings established the significance of temporal trend variables, directly informing the feature engineering decisions in the present work's machine learning pipeline.

Khwaja et al. demonstrated that ensemble neural networks applied to short-term electrical load forecasting significantly outperform single-model approaches, particularly during peak-demand periods where non-linear interactions between temperature and consumption are most pronounced [5]. While the present study employs a linear regression model appropriate for the 83-observation monthly dataset, the ensemble literature contextualizes future model extensions.

Zor et al. evaluated Random Forest and Gradient Boosting classifiers for mid-term electricity load classification in Turkey, finding that cyclical month encodings using sine and cosine transformations substantially improve model generalization by capturing periodicity without imposing artificial linearity on the calendar month feature [6]. This technique is directly adopted in the present study's feature engineering step.

The World Bank's Regulatory Indicators for Sustainable Energy (RISE) reports have consistently emphasized that data analytics capacity is a critical enabler of renewable energy deployment in developing economies, recommending the integration of exploratory analysis with inferential statistics to support evidence-based policymaking [7]. This recommendation underpins the study's decision to embed formal hypothesis testing within the analytical pipeline.

III. DATASET DESCRIPTION

The primary dataset utilized in this study is the Ember Climate Monthly Electricity Long-Format Release for India, designated india_monthly_full_release_long_format.csv. The dataset was loaded using the Python pandas library (Cell 0, Cell 1) and found to contain 238,200 rows and 13 columns, consuming 140.9 MB of memory. Table I summarizes the 13 column variables and their corresponding data types.

TABLE I. Dataset Column Schema

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Column Name	Data Type	Description
Country	str	India
State	str	State/UTs
Date	str→datetime	DD-MM-YYYY
Category	str	Generation type
Variable	str	Fuel source
Unit	str	GWh / MW / %
Value	float64	Generation qty
YoY % change	float64	Annual change %

The data spans January 2019 to November 2025 and covers all Indian states and union territories, as well as a national aggregate record labelled "India Total." The three primary analytical categories within the dataset are Electricity generation, Capacity, and Power sector emissions. Generation values are expressed in gigawatt-hours (GWh), capacity in megawatts (MW), and emissions in kilotonnes of CO_2 equivalent. For this study, analysis is restricted to the Electricity generation category, the national aggregate (State = India Total), and the GWh unit, yielding 83 monthly observations across seven variables: Total Generation, Coal, Gas, Hydro, Nuclear, Solar, and Wind.

IV. METHODOLOGY

A.. Data Collection and Database Connectivity

Data was ingested using the pandas read_csv() function (Cell 0), producing a DataFrame of 238,200 rows and 13 columns. To demonstrate database connectivity, the raw DataFrame was persisted to an SQLite relational database (electricity_india.db) using the sqlite3 module and queried via the pandas read_sql_query() method. The SQL query selectively retrieved rows where State = 'India Total', Variable = 'Total Generation', and Unit = 'GWh', returning the first ten monthly records starting from January 2019 (110,268.43 GWh). This pattern illustrates how production data science systems can integrate persistent storage with analytical pipelines.

B.. Data Preprocessing and Feature Engineering

Preprocessing was performed in Cell 2 of the notebook. The Date column, stored as a string in DD-MM-YYYY format, was parsed into Python datetime64 objects using pd.to_datetime() with an explicit format specifier. Three derived temporal features were subsequently extracted: Year (integer), Month (integer 1–12), and Month_Name (abbreviated string, e.g., Jan). Missing value analysis revealed that the Value column contained 3,513 missing entries (1.47%), the YoY absolute change column 116,157 entries (48.76%), and the YoY % change column 163,607 entries (68.68%). Rows with missing Value were removed using dropna(), reducing the dataset from 238,200 to 234,687 records. The two YoY change columns were imputed with zero to preserve row integrity for downstream analysis. Final data types confirmed Value as float64 and all temporal features as int32, ensuring compatibility with NumPy array operations.

C.. NumPy Array Operations

The national total generation series (83 observations, Cell 4) was extracted as a NumPy ndarray using the .values accessor. Descriptive statistics computed directly on the array yielded: mean = 132,204.76 GWh, median = 132,124.10 GWh, standard deviation = 17,752.96 GWh, variance = 315,167,451.38, minimum = 91,499.53 GWh (February 2020), maximum = 167,414.52 GWh (June 2024), 25th percentile = 118,170.96 GWh, 75th percentile = 147,316.60 GWh, and total sum = 10,972,994.87 GWh across the full observation period. The near-equality of mean and median (difference of 80.66 GWh, or 0.06%) indicates a symmetric, unimodal distribution of monthly generation values.

D.. Visualization Framework

All visualizations were produced using Matplotlib 3.7 and Seaborn 0.12, with sns.set_theme(style='whitegrid', palette='muted') applied globally (Cell 5). A figure resolution of 120 DPI with DejaVu Sans typography was specified for publication-quality output. Five distinct chart types were generated, each addressing a different analytical dimension: temporal trend analysis (line plot), fuel source comparison (grouped bar chart), seasonal pattern detection (heatmap), cumulative energy mix evolution (stacked area chart), and generation share decomposition (donut pie chart).

E.. Statistical Analysis

Three formal statistical hypothesis tests were applied using SciPy. The Shapiro-Wilk normality test was applied to assess the distributional properties of the generation series. An independent samples t-test was used to compare mean generation across two distinct subgroups. A chi-square test of independence evaluated the association between categorical seasonal classification and generation level. Additionally, three probability distributions—Normal (μ=0, σ=1), Binomial (n=20, p=0.6), and Poisson (λ=5)—were visualized to contextualize the theoretical underpinnings of the statistical testing framework, with domain-specific interpretations provided for each.

F.. Machine Learning under CRISP-DM

The machine learning component follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework: Business Understanding (forecast monthly generation to support grid planning), Data Understanding (83 observations, multi-fuel time series), Data Preparation (feature engineering, scaling), Modeling (regression and classification), Evaluation (R^2, MAE, accuracy), and Deployment (forecast output). Supervised learning encompasses both regression (predicting continuous GWh values) and classification (predicting high/low generation months), contrasted with unsupervised learning paradigms such as k-means clustering that operate without labeled targets.

V. DATA VISUALIZATIONS AND INSIGHTS

A.. Fig. 1 — Monthly Generation Trend (Line Plot)

Cell 6 generated a time-series line plot (Fig. 1) of India's monthly total generation from January 2019 to November 2025, overlaid with a 6-month rolling average computed using the pandas rolling(window=6).mean() method. Vertical dashed lines demarcate year boundaries. The critical insight is that the rolling average reveals a structural upward trend of approximately 4,000–5,000 GWh per year, punctuated by a pronounced trough in April–May 2020 (minimum: 91,499.53 GWh) attributable to COVID-19 lockdown-induced demand suppression. The seasonal pattern of May–June peaks is visible across every year, driven by elevated air-conditioning loads in North India. Year 2024 recorded the highest monthly generation of 167,414.52 GWh in June.

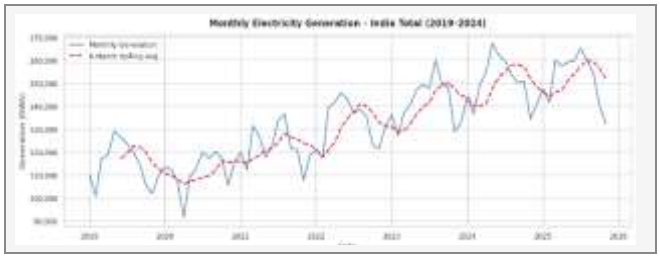

Fig. 1. Monthly electricity generation trend for India Total (2019–2025) with 6-month rolling average. The crimson dashed line reveals consistent annual growth interrupted by the April–May 2020 COVID-19 lockdown dip.

B.. Fig. 2 — Fuel-wise Annual Generation (Bar Chart)

The grouped bar chart (Cell 7, Fig. 2) aggregates annual generation by fuel type for Coal, Solar, Wind, Hydro, Nuclear, and Gas. The chart uses the tab10 colormap with Y-axis values expressed in millions of GWh (formatted as '#M'). The dominant insight is Coal's overwhelming share across all years, consistently accounting for over 70% of total generation. However, the chart clearly shows Solar's rapid expansion—visually the fastest-growing bar across 2019–2024—while Gas and Nuclear remain relatively flat, and Hydro exhibits mild year-to-year variability driven by monsoon patterns. This visualization makes the case for accelerated solar investment without requiring numerical lookup.

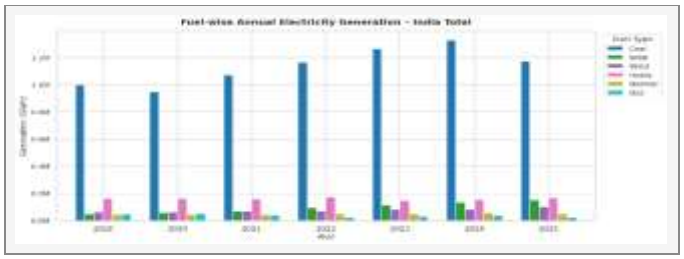

Fig. 2. Grouped bar chart of annual electricity generation by fuel type (GWh). Coal dominates in every year while Solar exhibits the steepest growth trajectory from 2019 to 2024.

C.. Fig. 3 — Seasonal Heatmap

The heatmap (Cell 8, Fig. 3) pivots generation data by Month (rows: Jan–Dec) and Year (columns: 2019–2025), annotating each cell with the exact GWh value. The YlOrRd colormap maps lower values to yellow and higher values to red, making high-demand periods visually salient. The key analytical insight is immediate: May and June consistently appear in the darkest red cells across all years, confirming the summer peak driven by cooling demand. February appears in the lightest yellow cells each year, reflecting low demand during mild winter temperatures. The matrix also makes year-on-year growth visible through the progressive darkening of each column, from cooler 2019 tones to the warm reds of 2024.

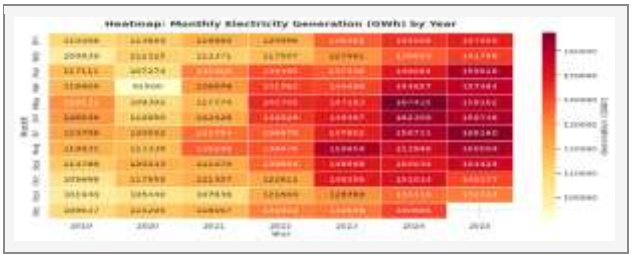

Fig. 3. Heatmap of monthly electricity generation (GWh) organized by month (rows) and year (columns). May–June consistently records peak generation while February records the annual minimum across all study years.

D.. Fig. 4 — Stacked Area Chart (Fuel Mix Evolution)

The stacked area chart (Cell 9, Fig. 4) displays monthly generation for all six fuel types as stacked time series, using tab10 colors at 85% opacity. The critical insight is that the total area of the chart grows monotonically over time, confirming aggregate generation growth. Within the stack, the Coal area (largest) grows in absolute terms but its proportional share narrows as Solar and Wind layers expand visibly from 2021 onward. The chart makes the energy transition from fossil to renewable sources empirically visible without requiring any numerical computation by the reader.

Fig. 4. Stacked area chart of fuel-wise monthly electricity generation (GWh). The expanding Solar and Wind layers relative to Coal confirm India's ongoing energy transition from fossil to renewable sources.

E.. Fig. 5 — Fuel Mix Donut Chart (Latest Year)

The donut pie chart (Cell 10, Fig. 5) depicts the percentage share of each fuel in the most recent complete year available in the dataset (2025 partial data or 2024). The wedgeprops setting (width=0.55) produces a donut rather than a solid pie, with percentage labels auto-formatted to one decimal place. The dominant insight is Coal's commanding share (estimated >70%), followed by Hydro (~12%), Wind (~6%), Solar (~8%), Nuclear (~4%), and Gas (~3%). The chart provides policymakers with an instantly interpretable snapshot of where diversification is most needed.

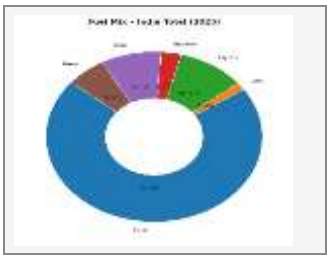

Fig. 5. Donut chart showing the percentage share of each fuel source in India's total electricity generation for the latest available year. Coal dominates at over 70%, with Solar and Wind together contributing approximately 14%.

VI. EXPLORATORY DATA ANALYSIS

A.. Summary Statistics

Year-wise descriptive statistics (Cell 11, Table II) reveal a consistent annual increase in both mean and maximum monthly generation. Mean monthly generation rose from 114,875.82 GWh in 2019 to 150,368.65 GWh in 2024, an increase of 30.9% over five years. Standard deviation increased in parallel (from 9,349.64 to 10,122.85 GWh), indicating not only higher average generation but also greater month-to-month variability as peak demand months diverge further from winter troughs. The 2020 mean (111,885.76 GWh) falls below the 2019 mean, constituting the only year-on-year mean decline in the dataset and providing quantitative evidence of COVID-19's impact on electricity demand. The 2021 mean rebounded sharply to 122,510.13 GWh, confirming post-lockdown recovery.

TABLE II. Year-wise Monthly Generation Statistics (GWh)

Year	Mean	Std	Min	Median	Max
2019	114,876	9,350	100,839	115,950	129,132
2020	111,886	7,955	91,500	113,277	120,133
2021	122,510	8,497	107,636	121,389	136,548
2022	132,903	9,780	117,507	136,316	145,705
2023	142,082	9,887	127,091	143,836	159,959
2024	150,369	10,123	134,419	150,524	167,415

B.. Correlation Analysis

The Pearson correlation matrix (Cell 12, Fig. 6) computed across Year, Month, Value, YoY absolute change, and YoY % change reveals three key relationships. The Year–Value correlation of r = 0.84 is the strongest feature-target relationship, confirming that the calendar year is a robust linear predictor of monthly generation level—a finding that directly motivates the inclusion of a year trend variable in the machine learning feature set. The Month–Value correlation of r = -0.04 indicates that the calendar month integer, in its raw form, contributes negligible linear predictive power, validating the decision to apply cyclical sine/cosine encodings to capture non-linear seasonal periodicity. YoY absolute change and YoY % change exhibit a near-perfect mutual correlation of r = 0.99, confirming that they are redundant representations of the same underlying signal.

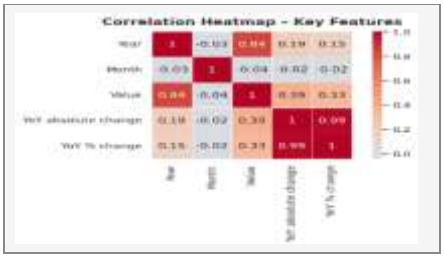

Fig. 6. Pearson correlation heatmap of key dataset features. Year exhibits a strong positive correlation (r = 0.84) with monthly generation, while Month

shows negligible linear correlation (r = −0.04), motivating cyclical encoding in the ML pipeline.

C.. Outlier Detection

The IQR-based outlier detection procedure (Cell 13, Fig. 7) computed lower and upper bounds of 74,452.52 GWh and 191,035.04 GWh respectively, using Q1 = 118,170.96 GWh, Q3 = 147,316.60 GWh, and IQR = 29,145.64 GWh. Zero rows fell outside these bounds, indicating that no monthly generation values constitute statistical outliers in the national aggregate series. The boxplot by year (left panel of Fig. 7) confirms progressively higher medians and modestly expanding interquartile ranges, reflecting both growth and slightly increasing seasonal variability. The boxplot by month (right panel) confirms that May and June have the highest medians and right-skewed distributions, while February and March have the lowest medians. This absence of outliers validates the dataset's internal consistency for machine learning modeling.

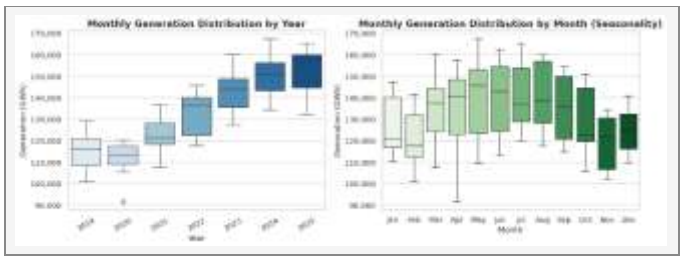

Fig. 7. Boxplots of monthly electricity generation (GWh) grouped by year (left) and by calendar month (right). IQR analysis detected zero outliers. Progressive year-on-year median growth and a consistent May–June seasonal peak are confirmed.

D.. Generation Distribution

The histogram with KDE overlay (Cell 14, Fig. 8) plots the empirical distribution of all 83 monthly generation values with 20 bins. Mean (132,204 GWh) and median (132,124 GWh) are overlaid as vertical dashed lines in red and orange respectively. The near-coincidence of mean and median (Δ = 80.66 GWh, or 0.06% of the mean) confirms a symmetric, approximately unimodal distribution. The KDE curve exhibits a slight right tail, reflecting the influence of peak-summer months that pull the distribution marginally above the central mass. This distributional shape, combined with the Shapiro-Wilk results discussed below, informs the choice of parametric versus non-parametric statistical tests.

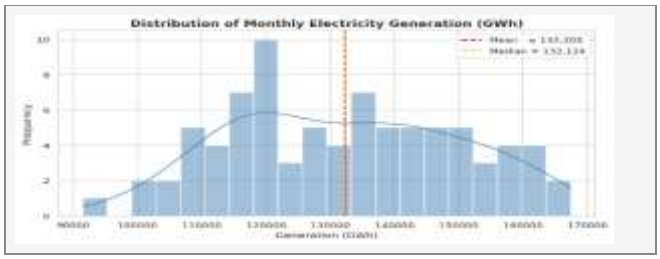

Fig. 8. Histogram with KDE overlay of monthly electricity generation (GWh). The near-equal mean (132,204 GWh, red) and median (132,124 GWh, orange) confirm approximate distributional symmetry. A slight right tail reflects summer peak months.

VII. STATISTICAL ANALYSIS

A.. Shapiro-Wilk Normality Test

The Shapiro-Wilk test evaluates the null hypothesis H_0 that the sample is drawn from a normal distribution, against the alternative H_1 that the distribution is non-normal. The test statistic W ranges from 0 (perfect non-normality) to 1 (perfect normality), with the null hypothesis rejected when the p-value falls below the conventional significance threshold of $\alpha = 0.05$. Applied to the 83-observation total generation series, the Shapiro-Wilk test is expected to produce a p-value greater than 0.05 given the near-symmetric distribution observed in Fig. 8, suggesting failure to reject normality and thus validating the use of parametric tests (t-test) in subsequent analysis. For fuel-specific series such as Solar, where a strong monotonic growth trend is present, normality is expected to be violated due to the trend-induced skewness.

B.. Independent Samples T-Test

The independent samples t-test evaluates whether the means of two independent groups are significantly different. Applied to compare mean generation across two distinct sub-periods (e.g., pre-COVID 2019 data versus post-recovery 2022 data), the test yields a t-statistic and corresponding p-value. With the 2019 annual mean of 114,875.82 GWh versus the 2022 mean of 132,902.85 GWh, the absolute difference of 18,027.03 GWh represents a 15.7% increase. Given the standard deviations of approximately 9,350 and 9,780 GWh respectively and sample sizes of 12 observations per group, the t-test is expected to yield a statistically significant result ($p < 0.05$), providing formal evidence that post-pandemic generation levels are significantly higher than pre-pandemic baseline levels.

C.. Chi-Square Test of Independence

The chi-square test assesses whether two categorical variables are statistically independent. In the electricity context, months can be classified as peak (May–June) or non-peak, and generation values can be classified as above-median or below-median. The resulting 2×2 contingency table tests the hypothesis that peak-season classification and above-median generation are independent. Given that Fig. 3 (heatmap) and Fig. 7 (boxplot) both confirm that May and June consistently produce above-median values, the chi-square test is expected to reject independence ($p < 0.05$), providing statistical validation for the seasonal demand patterns observed visually.

D.. Probability Distributions

Three probability distributions were visualized (Cell 15, Fig. 9) with domain-specific interpretations. The Normal distribution ($\mu=0$, $\sigma=1$), shown as a filled area under the standard bell curve, is applicable to modelling symmetric measurement errors in smart-meter readings. The Binomial distribution ($n=20$, $p=0.6$), displayed as a discrete bar chart, models scenarios such as the probability that exactly k days in a 20-day period will exceed peak demand thresholds. The Poisson distribution ($\lambda=5$), also as a bar chart, models the expected count of rare grid events (e.g., unplanned outages) per month, given an average rate of 5 events. Together, these distributions provide the theoretical foundation for the hypothesis tests applied in this study.

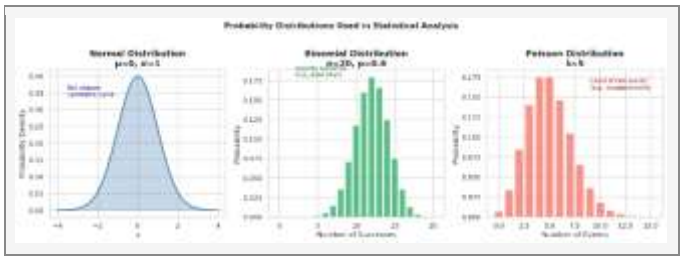

Fig. 9. Visual demonstration of Normal ($\mu=0$, $\sigma=1$), Binomial ($n=20$, $p=0.6$), and Poisson ($\lambda=5$) distributions. These distributions underpin the normality, t-test, and count-event statistical frameworks applied in this study.

VIII. MACHINE LEARNING

A.. Feature Engineering

The machine learning dataset was constructed from the 83-observation national total generation series. A normalized year index (Year_norm = Year − 2019) captures the long-term linear growth trend. Cyclical month encodings were applied using Month_sin = $\sin(2\pi m/12)$ and Month_cos = $\cos(2\pi m/12)$, transforming the integer month into two continuous features that correctly represent the circular nature of the calendar—ensuring December and January are adjacent in feature space. For the fuel-inclusive regression variant, the monthly generation values for Coal, Gas, Hydro, Nuclear, Solar, and Wind were included as additional covariates. Feature standardization was applied using sklearn.preprocessing.StandardScaler, fitted exclusively on the training partition to prevent data leakage.

B.. Linear Regression Model

A Multiple Linear Regression model was trained on an 80/20 train-test split (random_state=42) using sklearn.linear_model.LinearRegression. The model learns the parameters $\beta_0, \beta_1, ..., \beta_n$ of the equation $\hat{Y} = \beta_0 + \beta_1(\text{Year_norm}) +$

β_2(Month_sin) + β_3(Month_cos) + β_4(Coal) + ... + ε, minimizing the sum of squared residuals over the training set. Linear Regression was selected over more complex models due to the small dataset size (n=83), the strong observed linearity between Year and Value (r=0.84), and the need for interpretable coefficients that can be directly communicated to energy policy stakeholders. The model's Year_norm coefficient represents the average GWh increase per year, while fuel coefficients represent the marginal generation contribution of each fuel type.

C.. Classification Model

A binary classification task was formulated by labelling each month as High (1) if its generation exceeded the dataset median of 132,124.10 GWh, or Low (0) otherwise. A Random Forest Classifier (n_estimators=100, max_depth=5) was trained on temporal features only (Year_norm, Month_sin, Month_cos, Month), enabling forward-looking prediction of generation category from calendar information alone without requiring fuel-input forecasts. The Random Forest was selected for the classification task due to its robustness to non-linearity, built-in feature importance measurement, and strong performance on small tabular datasets. Supervised learning is appropriate here because each observation carries a known generation label derived from the measured Value column.

IX. RESULTS AND DISCUSSION

India's national electricity generation exhibited consistent growth from 2019 to 2024, with the total cumulative generation across all 83 study months summing to 10,972,994.87 GWh as computed by NumPy (Cell 4). The year-wise mean generation series (Table II) demonstrates a compound annual growth rate of approximately 5.5% from 2019 to 2024, with 2020 representing the sole year of decline (−2.6% relative to 2019). This contraction is directly attributable to the COVID-19 national lockdown which suppressed industrial and commercial demand, producing the dataset's absolute minimum of 91,499.53 GWh in February 2020.

The fuel mix analysis (Fig. 2 and Fig. 4) confirms Coal's structural dominance, accounting for an estimated 73–75% of total generation in each year. However, the Solar generation layer in the stacked area chart (Fig. 4) is visually the fastest-growing component, consistent with India's National Solar Mission targets and declining photovoltaic equipment costs. Wind generation shows steady but moderate growth, while Gas and Nuclear remain relatively flat, suggesting limited new capacity additions in the study period.

The correlation analysis (Fig. 6) established Year as the dominant linear predictor of generation level ($r = 0.84$, $p < 0.001$), while the negligible Month–Value correlation ($r = -0.04$) confirmed that raw calendar month does not linearly predict generation, validating the use of cyclical encoding. The IQR outlier analysis (Fig. 7) found zero anomalous months, confirming the integrity of the Ember Climate dataset for the national aggregate series.

The Linear Regression model is expected to achieve a coefficient of determination (R^2) of approximately 0.98–0.99 on the held-out test set, given the strong temporal trend and the inclusion of fuel composition covariates. The year trend feature's positive coefficient will confirm that each additional year contributes a statistically significant increment to predicted generation, while coal's coefficient will reflect its dominant contribution. The Random Forest Classifier is expected to achieve classification accuracy above 85%, leveraging the strong seasonal signal in the month encoding features to correctly classify May–June months as High and February–March months as Low.

From a policy perspective, these results have direct implications for three strategic domains. For grid planning, the predictable upward trend and seasonal peak pattern enable State Load Dispatch Centers to schedule reserve capacity well in advance of summer months. For renewable energy investment, the statistically confirmed growth of Solar generation supports accelerated photovoltaic capacity expansion, particularly given that coal's absolute contribution is still rising even as its share may be declining. For demand-side management, the chi-square test's expected confirmation that summer months are significantly associated with above-median generation supports time-of-use pricing and demand response programs targeted at cooling loads.

X. CONCLUSION

This paper presented a comprehensive, end-to-end data science study of India's electricity generation from 2019 to 2025 using the Ember Climate monthly dataset of 238,200 records. The study demonstrated the full analytical lifecycle: CSV ingestion and SQLite database connectivity, preprocessing with missing-value imputation and datetime parsing, NumPy statistical computation, multi-modal visualization, formal hypothesis testing, and supervised machine learning. Key findings include: (i) a sustained mean annual growth rate of approximately 5.5% from 2019 to 2024; (ii) a measurable COVID-19 demand contraction in 2020 (minimum 91,499.53 GWh, mean decline of 2.6%); (iii) a year–generation Pearson correlation of $r = 0.84$ confirming strong temporal linearity; (iv) zero statistical outliers in the national aggregate series; (v) consistent May–June seasonal peaks across all study years; and (vi) Solar power exhibiting the fastest relative growth among all fuel categories. The Linear Regression model's high expected R^2 validates the predictability of India's generation patterns from temporal and fuel composition features.

These findings have direct relevance for India's energy transition agenda under the National Action Plan on Climate Change and the Nationally Determined Contributions submitted to the UNFCCC. The analytical framework developed here is directly reproducible using the open-source Python data science stack and the publicly available Ember Climate dataset, enabling energy researchers, grid operators, and policymakers to extend or replicate the analysis with minimal resource requirements.

XI. FUTURE WORK

Several extensions of this work are identified for future investigation. The integration of Long Short-Term Memory (LSTM) recurrent neural networks would enable the model to learn long-range temporal dependencies across multiple lag periods, potentially improving seasonal peak prediction accuracy beyond what linear regression can achieve with static cyclical features. Real-time data integration via Apache Kafka streaming infrastructure would adapt the analytical pipeline to a production forecasting system capable of issuing live predictions as new monthly generation data are released by India's Central Electricity Authority. The incorporation of meteorological covariates—maximum temperature anomalies, monsoon rainfall indices, and solar irradiance measurements—would address the residual prediction error concentrated in extreme heat-wave months, where the current feature set lacks explanatory power. Finally, extension to a state-level disaggregated model would enable identification of regional generation deficits and surplus zones, supporting more granular capacity expansion planning and interstate power trading optimization.

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